

UGINO, Russian Federation

WMO: 310160

Lat: 58.633N

Lon: 128.500E

Elev: 756

StdP: 14.30

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-55.1	-51.0	-57.8	0.2	-54.8	-53.8	0.2	-50.7	7.5	-7.1	6.7	-9.9	0.8	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.0	85.6	66.9	81.7	64.8	78.1	63.0	69.2	80.1	67.2	77.6	65.2	75.1	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	96.4	74.2	63.5	90.0	72.0	61.4	83.8	70.0	33.9	79.9	32.3	77.4	30.6	75.3	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	7.6	6.2	-58.6	92.4	6.6	4.1	-63.3	95.4	-67.1	97.8	-70.8	100.2	-75.6	103.2		
(5)			-58.2	73.9	6.8	3.5	-63.1	76.4	-67.1	78.5	-70.9	80.4	-75.8	83.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	-29.1	-18.3	4.9	27.0	44.7	59.2	64.2	59.7	43.6	22.5	-9.4	-24.8		
	DBStd	34.94	13.89	12.74	13.16	9.93	7.58	6.58	6.20	6.50	7.31	10.61	13.78	14.76		
	HDD50	11833	2453	1913	1399	691	196	9	0	8	213	852	1782	2317		
	HDD65	16329	2918	2333	1864	1141	629	192	92	188	641	1317	2232	2782		
	CDD50	1086	0	0	0	0	32	286	439	307	22	0	0	0		
	CDD65	108	0	0	0	0	0	19	66	23	0	0	0	0		
	CDH74	1726	0	0	0	0	25	453	909	335	4	0	0	0		
	CDH80	497	0	0	0	0	4	95	322	75	1	0	0	0		
	WSAvg	2.1	1.3	1.7	2.2	2.8	3.0	2.4	2.2	2.1	2.0	2.4	1.8	1.5		
	PrecAvg	19.8	0.7	0.5	0.6	1.0	2.0	2.4	3.3	3.0	2.6	1.8	1.1	0.8		
(15) Precipitation	PrecMax	25.3	1.3	0.8	1.2	2.4	3.1	4.4	6.0	5.9	5.0	4.0	2.5	1.5		
	PrecMin	15.6	0.3	0.2	0.3	0.4	0.9	1.0	1.2	0.7	0.9	0.9	0.4	0.5		
	PrecStd	2.7	0.2	0.2	0.2	0.5	0.6	0.8	1.1	1.2	0.9	0.7	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	18.9	44.1	59.5	77.0	87.7	92.8	87.4	71.3	51.1	37.4	21.9		
		MCWB	8.2	17.3	34.8	45.7	56.2	66.1	69.7	67.6	56.9	41.8	33.4	19.7		
	2%	DB	1.9	12.6	36.0	52.4	70.8	82.6	88.0	82.1	65.6	44.4	24.9	12.5		
		MCWB	0.9	11.4	30.4	40.4	52.8	63.5	68.1	65.7	53.6	38.6	21.9	11.7		
	5%	DB	-3.2	7.0	31.6	48.5	65.9	79.4	83.6	77.8	60.8	39.5	16.0	3.7		
		MCWB	-3.9	5.8	27.3	37.9	50.3	61.6	66.6	64.1	51.5	35.4	14.7	3.1		
	10%	DB	-8.7	1.4	27.2	44.6	61.5	76.2	79.2	74.0	56.5	36.1	10.1	-4.0		
		MCWB	-9.2	0.3	23.8	36.1	47.9	60.3	64.8	62.6	49.1	33.2	9.2	-4.5		
	0.4%	WB	8.7	17.6	35.5	46.0	57.6	69.2	72.7	70.2	59.1	43.0	32.8	20.1		
		MCDB	10.1	19.3	42.0	59.4	74.0	79.2	87.2	81.4	66.5	49.4	36.9	22.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.1	11.4	30.8	41.2	54.3	65.7	70.2	67.9	55.3	39.8	22.6	11.6		
		MCDB	1.8	12.5	34.9	50.9	68.2	77.6	82.2	77.2	62.7	44.0	24.8	12.6		
	5%	WB	-3.8	6.0	27.3	38.4	51.7	63.7	68.3	65.7	52.7	36.0	14.7	3.1		
		MCDB	-3.1	7.1	31.4	47.4	62.5	76.3	79.4	75.0	59.4	38.8	16.0	3.8		
	10%	WB	-9.1	0.6	23.6	36.2	49.1	61.6	66.6	63.8	50.0	33.2	9.1	-4.3		
		MCDB	-8.6	1.4	27.2	44.0	59.4	73.2	76.9	72.9	55.5	36.0	10.0	-3.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	23.9	31.3	26.9	24.3	25.9	24.0	23.4	20.1	15.2	17.5	14.9		
		MCDBR	17.9	26.2	30.1	29.9	35.0	33.7	33.2	31.0	30.3	17.2	19.6	18.8		
		MCWBR	17.6	24.8	24.9	19.1	19.5	17.3	16.5	16.1	18.7	11.3	17.8	18.3		
		MCDBR	18.1	25.9	29.0	28.8	30.8	28.7	26.6	26.3	26.3	15.6	19.2	18.7		
	5% WB	MCWBR	17.6	24.6	23.9	18.8	18.2	16.3	14.9	15.3	17.1	11.0	17.5	18.2		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.282	0.318	0.393	0.466	0.438	0.479	0.430	0.349	0.308	0.256	0.229		
	taud		2.006	2.027	2.013	1.925	1.981	2.213	2.113	2.239	2.416	2.240	2.027	1.948		
	Ebn at Noon		176	228	255	252	242	253	238	242	249	220	167	137		
	Edh at Noon		17	28	38	50	51	41	45	37	26	23	16	13		
(44) All-Sky Solar Radiation	RadAvg		202	530	1080	1616	1527	1742	1605	1265	797	413	253	124		
	RadStd		13	30	41	98	130	124	120	103	64	43	16	7		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+1.51	N/A	N/A	N/A	N/A

Nomenclature: See separate page